

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First / Second Semester B.Tech (CBCS) Degree Examinations
 Summer Semester July 2023
CV1001 - 1 – ELEMENTS OF CIVIL ENGINEERING



Duration: 3 Hours

Note:

*Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.*

*Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. _____ deals with a detailed study of planning, design, construction and maintenance of different types of roadways, railways, airports and runways, harbours, bridges and tunnels.
 A) Environmental Engineering B) Transportation Engineering
 C) Geotechnical Engineering D) Structural Engineering
2. _____ provides methods and facilities for waste water management, water and air purification, waste disposal and recycling, and other purposes pertaining to human health and benefit.
 A) Irrigation Engineering B) Hydraulics
 C) Geotechnical Engineering D) Environmental Engineering
3. Which of the following is not a basic idealization of Engineering Mechanics?
 A) Particle B) Continuum
 C) Self Weight D) Rigid Body
4. _____ is an idealized body which may have finite or negligible mass and whose size and shape can be ignored without sacrificing the accuracy.
 A) Continuum B) Particle
 C) Rigid Body D) Point Force
5. If the arm of the couple is doubled, its moment will be _____.
 A) doubled B) remain the same
 C) halved D) zero
6. Which of the following conditions changes the effect of couple?
 A) Addition of another couple of same magnitude B) Rotation of couple by some angle in its plane
 C) Shifting of couple to a parallel plane D) Shifting of couple to a new position in its plane
7. Moment of Couple does not depend on _____.
 A) Arm of the couple B) Moment centre
 C) Moment centre and Arm of the couple D) All of these
8. Two couples are said to be equivalent if their _____.
 A) Rotation is same B) Magnitudes are same
 C) Moments are equal D) Signs are equal
9. Hinged supports offer _____ and _____ reactions.
 A) Moment, Couple B) Moment, Horizontal
 C) Moment, Vertical D) Vertical, Horizontal
10. Which of the following are statically determinate beams?
 A) Cantilever, overhanging and simply supported beam. B) Cantilever, overhanging and fixed beam
 C) Fixed, continuous and propped cantilever beam. D) Fixed, continuous and simply supported beam
11. In a simply supported beam, moment at the support _____.
 A) is zero, if it does not carry couple at the end B) is zero, if the beam has uniformly distributed load only
 C) is zero, if the beam has concentrated loads only D) may or may not be zero

12. _____ is a horizontal structural member subjected to transverse loads perpendicular to its axis.
 A) Truss B) Column
 C) Strut C) Beam
13. The magnitude of limiting friction bears a constant ratio to _____ is called coefficient of friction.
 A) Frictional Resistance B) Normal Reaction
 C) Wet Friction D) Dry friction
14. The frictional resistance depends upon the _____ of the surface.
 A) Area B) Diameter of the specimen
 C) Roughness D) Moisture Content
15. Frictional Resistance acts in a direction _____ to the motion of the body.
 A) Same B) Opposite
 C) Parallel D) Non-parallel
16. The frictional resistance is _____ of the area of contact between two bodies.
 A) Condition B) Dependent
 C) Nature D) Independent
17. On simplification of the loading system, the net force acts at the _____ of the loading body.
 A) Centroid B) The central axis
 C) The mid-point D) Inertial point
18. Density is best given by _____.
 A) Addition of mass and density B) Product of volume and density
 B) Ratio of mass to Volume D) Subtraction of mass and density
19. Which of the following laminas do not have centroid at its geometrical centre?
 A) Centre of gravity B) Right angled triangle
 C) Circle D) None of these
20. What is the vertical centroidal distance from base of right angled triangle of base 20 cm and side 40 cm?
 A) 13.33 cm B) 13.28 cm
 C) 19.36 cm D) 38.72 cm

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) (i) Explain the scope of following fields of Civil Engineering:
 Surveying, Construction Technology.
 (ii) Define force. Explain any two systems of forces. 4+4 L*2 1 1
 b) A bracket section is shown in Figure 1(b). Determine the magnitude, direction and position of resultant with respect to 'A'. 8 L3 2 2
2. a) (i) Explain the concept of resolution and composition of the forces.
 (ii) Reduce the force acting at 'A' into a system of force given in Figure 2(a) and couple at point 'O'. 4+4 L3 1 2
 b) Determine the forces in the wires shown in Figure 2(b). 8 3 2 1
3. a) Find the resultant of the force system shown in Figure 3(a) with respect to point O. 8 L3 1 1
 b) (i) State and prove Varignon's theorem.
 (ii) Distinguish between moment and couple. 6+2 L2 2 3

Unit – II

4. a) Calculate the support reactions for the Simply Supported Beam shown in Figure 4(a). 8 L3 3 1
 b) Two blocks A and B of mass 30 kg and 40 kg, respectively, are connected by means of a bar placed on an inclined plane as shown in Figure 4(b). If $\mu = 0.25$, calculate the value of angle of inclination required for the motion of the block to impend. 8 L3 4 1

5. a) Calculate the support reactions for the cantilever beam shown in Figure 5(a).
 b) (i) State any four Coulomb's laws of dry-friction.
 (ii) Write a note on types of friction.

8	L2	3	2
4+4	L3	4	1

6. a) (i) Explain roller support and hinged support with sketches.
 (ii) With examples explain the statically determinate and statically indeterminate beams.
 b) A uniform ladder 4 m long weighs 200 N. It is placed against a wall making an angle of 60° with the floor. The coefficient of friction between the wall and the ladder is 0.25 and that between the ground and the ladder is 0.35. The ladder in addition to its own weight, has to support a man of 1000 N at the top corner of the ladder. Draw the FBD of the ladder and determine the horizontal force P to be applied to the ladder at the ground level to prevent slipping.

4+4	L2	3	1
8	L3	4	3

Unit – III

7. a) Deduce the moment of inertia of rectangular lamina with respect to its centroidal axis by the method of integration.
 b) Determine the centroid of the shaded lamina shown in Figure 7(b) with respect to point 'O'.
 8. a) Deduce the centroid of semicircular lamina by the method of integration.
 b) Determine the moment of inertia about the centroidal axes for the lamina shown in Figure 8(b).

6	L2	5	1
10	L3	5	2
6	L2	5	1
10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

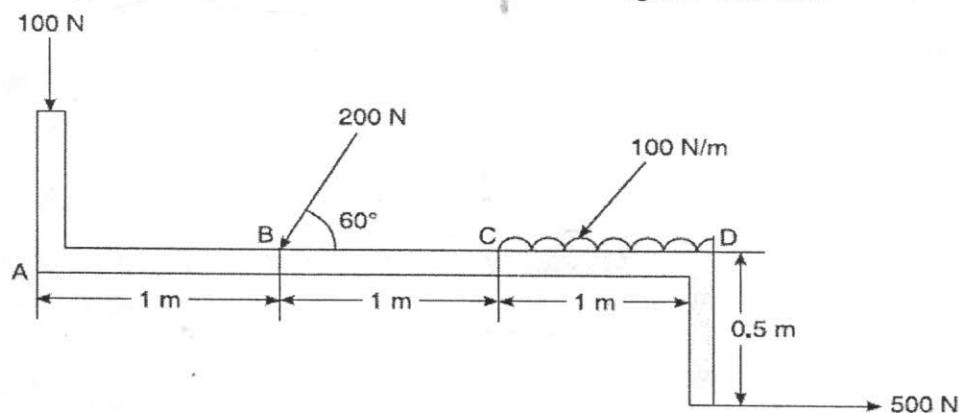


Fig. 1(b)

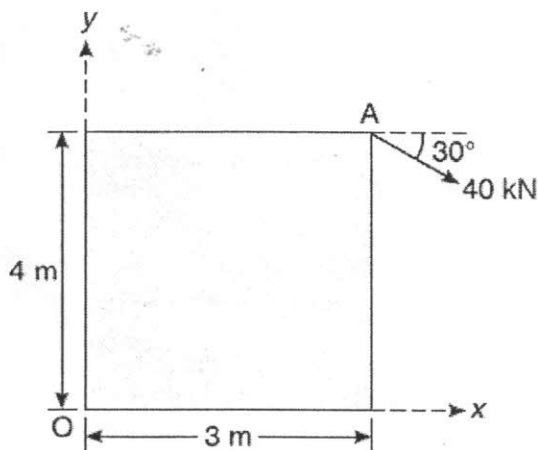


Fig. 2(a)

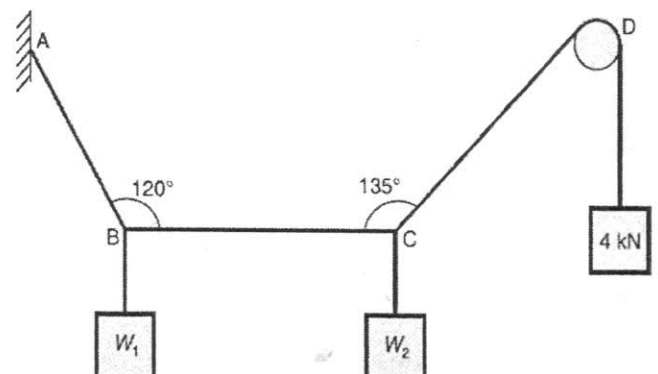


Fig. 2(b)

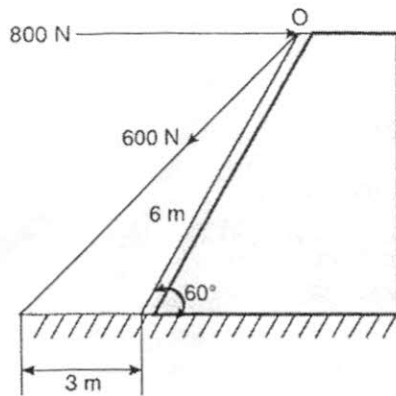


Fig. 3(a)

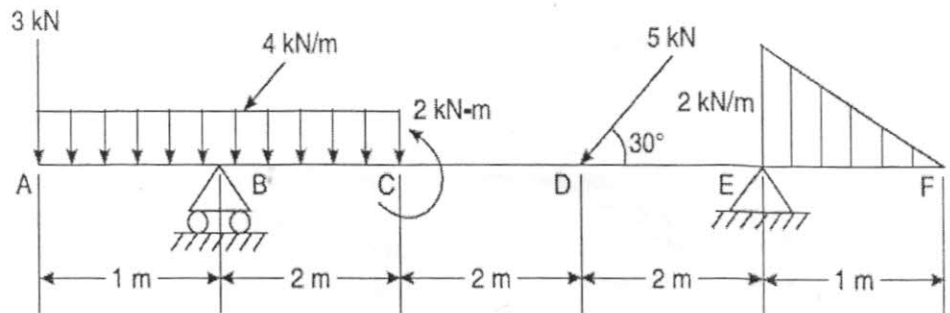


Fig. 4(a)

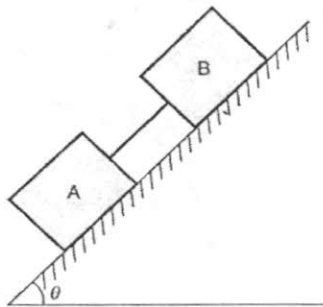


Fig. 4(b)

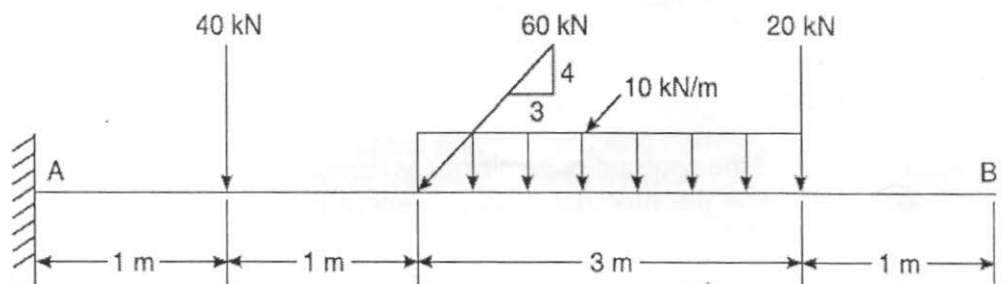


Fig. 5(a)

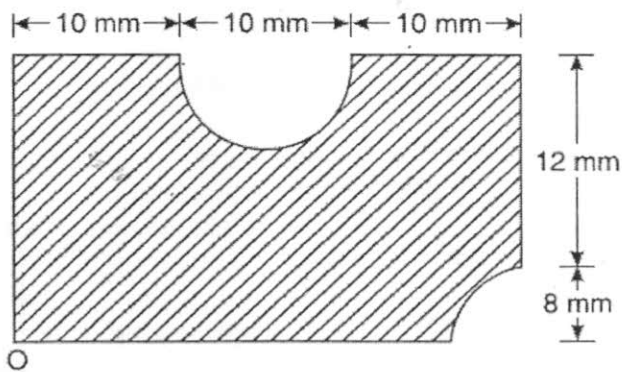


Fig. 7(b)

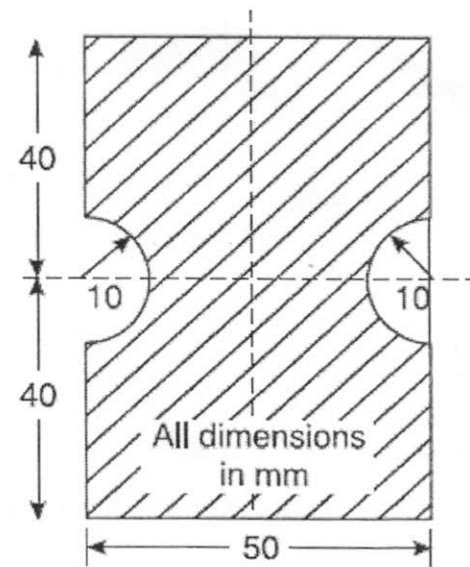


Fig. 8(b)
